THERMAL IMAGING-BASED FAULT DETECTION IN AN INDUCTION MOTOR USING MACHINE LEARNING TECHNIQUES

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Introduction

- Infrared ray is an electromagnetic wave and the range of infrared rays lies between human eyes perception and portions of the microwaves of an electromagnetic spectrum
- Infrared Thermography (IRT) is a method of measuring the temperature of an object by detecting the infrared radiations emitted from the object's surface
- In thermal imaging, thermal cameras convert these infrared radiations into the visible Images
- Thermal Imaging is used to detect these temperature differences
- As components wear away, their resistance increases, causing an increase in heat, and hence faults occur
- Forward Looking InfraRed (FLIR) thermal cameras are mostly used for capturing thermal images



Introduction (Cont'd)

- In electrical equipment detecting the external faults is easier, but the internal faults of the equipment depend on the varying temperature
- In an induction motor, several faults such as rotor fault, bearing fault, short circuit in stator windings, cooling fan failures, etc. can occur
- All these faults can be found accurately with efficient usage of thermal images
- This work mainly focuses on internal faults present in an induction motor and detecting the faults using machine learning classifiers

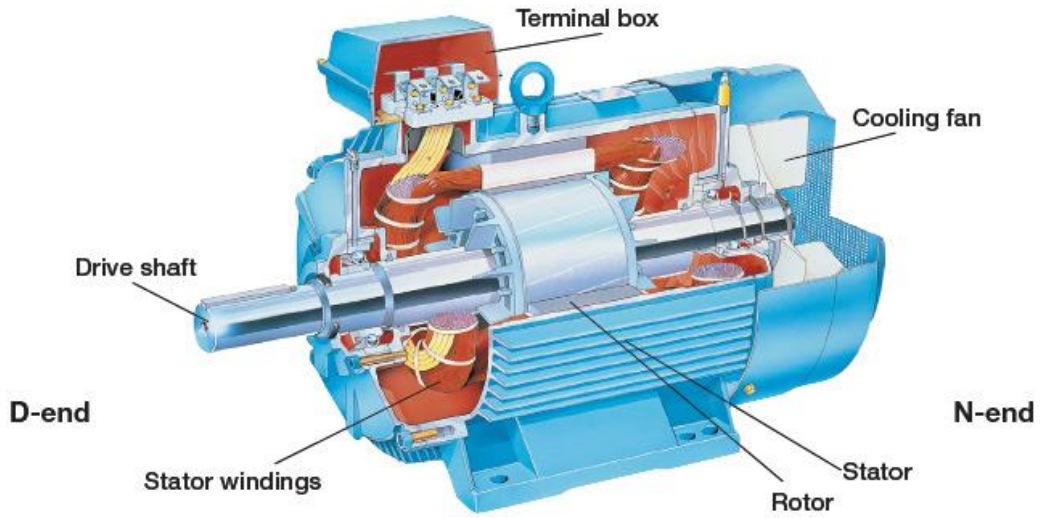


Image Source: Google

Literature review

Authors (Year)	Title of the paper	Key contributions /Inference
W. Yongqing et.al. (2017)	The temperature measurement technology of infrared thermal imaging and its applications review	▪ Analysis of Objects using Infrared thermal thermometer (FLIR-T420) ▪ Used in Military medical, electric power, construction, agriculture, Industrial fields for detecting faults
A. Choudhary, et.al. (2018)	Bearing fault diagnosis of induction motor using Thermal imaging	▪ Classification of Images is done using 2D-DWT, Mahalanobis distance and SVM Classifier for fault detection. ▪ Used in Three phase Induction Motor
Sambhi, Shilpa. (2018).	Thermal imaging technology for predictive maintenance of electrical installation in Manufacturing plant	▪ Monitoring performance of electrical motor using Predictive Maintenance. ▪ Comparing the operating temperature of similar equipment to find out faulty ones
O. Janssens et.al. (2019)	Thermal imaging and vibration-based multi-sensor fault detection for rotating machinery	▪ Two datasets containing vibration measurements and thermal imaging data , faults induced in a rotating setup and ML algorithms are used for preventive maintenance
J. Du et.al. (2019)	Application of infrared thermal imaging technology in fault diagnosis of mine car wheels	▪ Images are Grayscaled and binarized. Region growing algorithm is used. ▪ Wheel faults in underground Mine cars



Literature review (Cont'd)

Authors (Year)	Title of the paper	Key contributions /Inference
M. Najafi et.al. (2020)	Fault diagnosis of electrical equipment through thermal imaging and interpretable machine learning applied on a newly-introduced dataset	▪ Datasets consisting of thermal Images of an Induction Motor and Transformer and Classifying images based on hot and cold conditions ▪ Segmentation to find ROI and feature extraction and detecting faults using ML algorithms like Adaboost, Random forest and decision tree. ▪ Proposed Method gives the accuracy of 93.8%
A. A. Azeez et.al. (2020)	Thermal imaging fault detection for rolling element bearings	▪ Video recording of 10 mins using FLIR thermal Imaging camera, MATLAB. ▪ Faults Considered are No fault, Line fault, small circle fault, double line fault and large circle fault in rolling element bearings
A. Choudhary et.al. (2021)	Convolutional neural network based bearing fault diagnosis of rotating machine using thermal images	▪ Comparison based on classification performance using ANN and CNN. ▪ Rolling-element bearing in six conditions have been considered (one healthy and five faulty conditions)
A. Glowacz (2021)	Fault diagnosis of electric impact drills using thermal imaging	▪ BCAoID (Binarized Common Areas of Image Differences), KNN, and Backpropagation NN. ▪ Fault Diagnosis of healthy Electric Impact Drills, EID with faulty fan EID with the damaged gear train

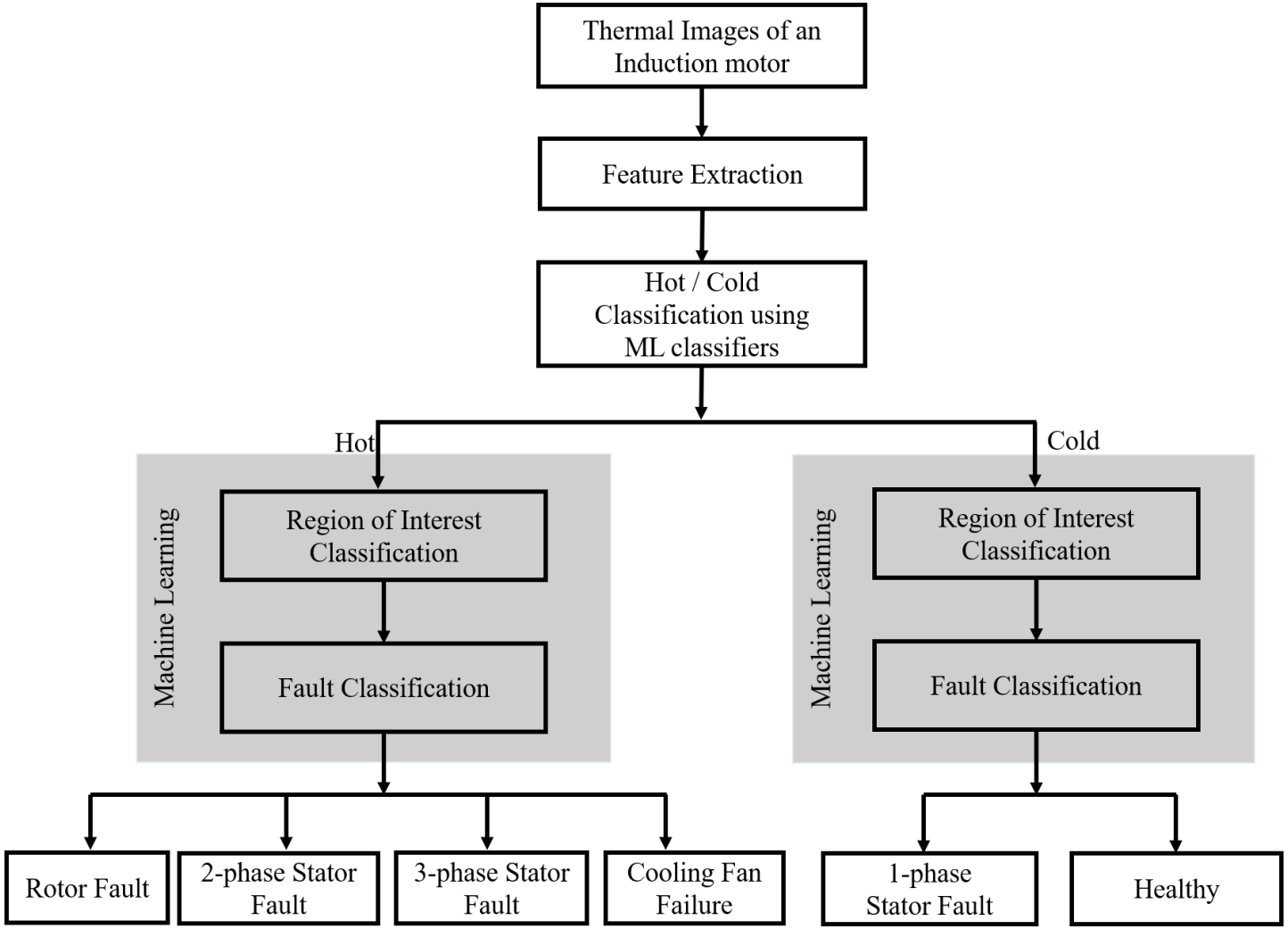
Literature review (Cont'd)

Authors (Year)	Title of the paper	Key contributions /Inference
A. Mehta et.al. (2021)	Machine learning-based fault diagnosis of self-aligning bearings for rotating machinery using infrared thermography	▪ Comparison of deep learning approaches such as ANN and CNN was discussed ▪ Principal Component Analysis (PCA) was used to select the features. ▪ Linear Discriminant Analysis (LDA), k-nearest neighbors (KNN), and Support Vector Machines (SVM) were used to categorize the faults
E. Resendiz-Ochoa et.al. (2021)	Broken Rotor Bar Failures Diagnosis with Supervised Learning and Infrared Thermography	▪ Three fault conditions are considered to find rotor bar faults using thermal images. (Half, One, and two) ▪ K-nearest neighbors (KNN) was used to detect and classify faults
R. Mehra (2022)	Thermal Imaging-Based Fault Diagnosis of Electronics Circuit Boards	▪ Arduino Uno and Raspberry Pi ▪ Principal Component Analysis (PCA) for dimensionality reduction ▪ Support vector machine (SVM) for fault detection
T. Mian et.al. (2022)	Multi-Sensor Fault Diagnosis for Misalignment and Unbalance Detection Using Machine Learning	▪ Misalignment and unbalanced detection in rotating machines ▪ Naive Bayes (NV), k-Nearest Neighbor (k-NN), and Support Vector Machine (SVM) for detecting faults

Objective

Using thermal imaging, the objective is to detect specific internal faults present in an induction motor by adopting machine learning classifiers

Proposed Methodology



Dataset

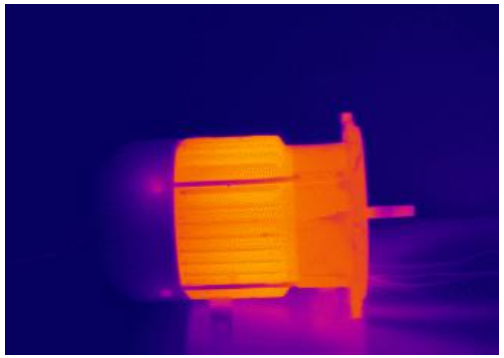
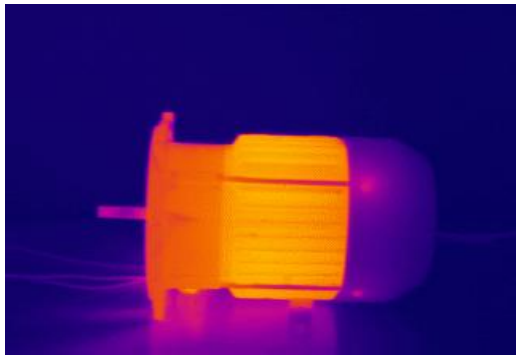
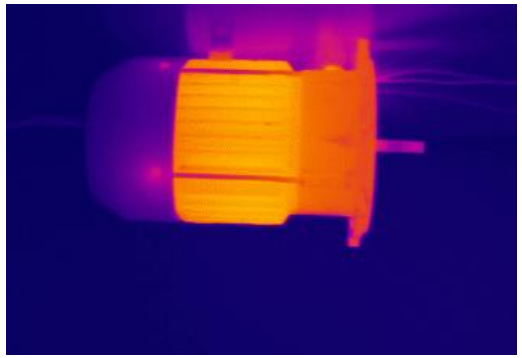
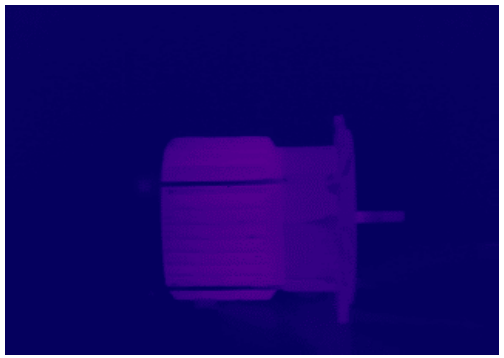
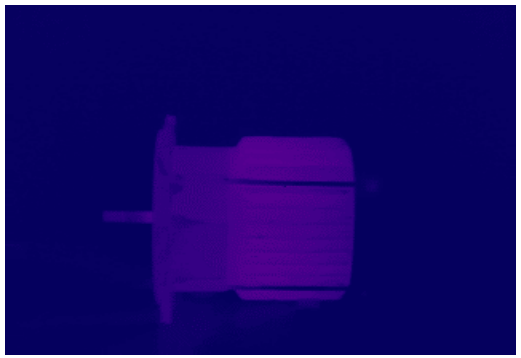
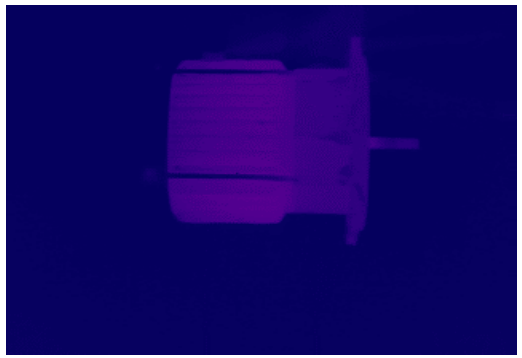
- The Mendeley dataset [26] is used which comprises infrared thermal images of induction motor.
- The thermal images comprise healthy and faulty conditions.
- The dataset has a total of 540 thermal images which includes faults in short circuits in stator windings, rotor, cooling fan, and healthy conditions.
- The healthy condition is treated as normal and the rest of the cases are treated as faulty
- Data Augmentation techniques are used
  - Horizontal flipping
  - Vertical flipping

Specifications of an induction motor	
Phase	3 Phase
Frequency	50 Hz
Power	1.1 KW
Voltage	220 / 380 V
Input Current	5 A
Speed	2800 RPM



Dataset (Cont'd)

- Sample thermal images

	Usual	Horizontally Flipped	Vertically Flipped
Hot			
Cold			

Implementation

Image Details

- The thermal Images present in the dataset are 3-D Images
- Consists of RGB (Red, Green, Blue) colors
- The dimensions of the Images are given by,

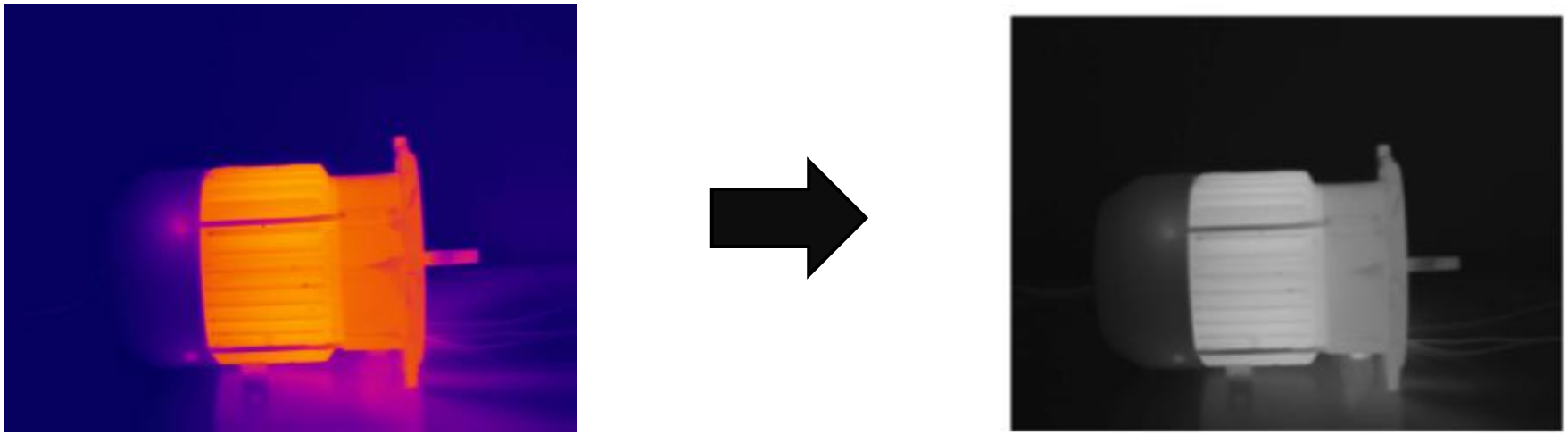
  (Height * Width * No. of channels) = (240 * 320 * 3)
- The datatype of the Images is “uint8”



Implementation (Cont'd)

RGB to gray scale images

- RGB → Gray Scale Images (3D → 2D)
- Conversion to gray scale converts the image into single layer image



Implementation (Cont'd)

Feature Extraction

- The histogram or first-order features and textural or second order gray level cooccurrence matrix (GLCM) features are as shown,

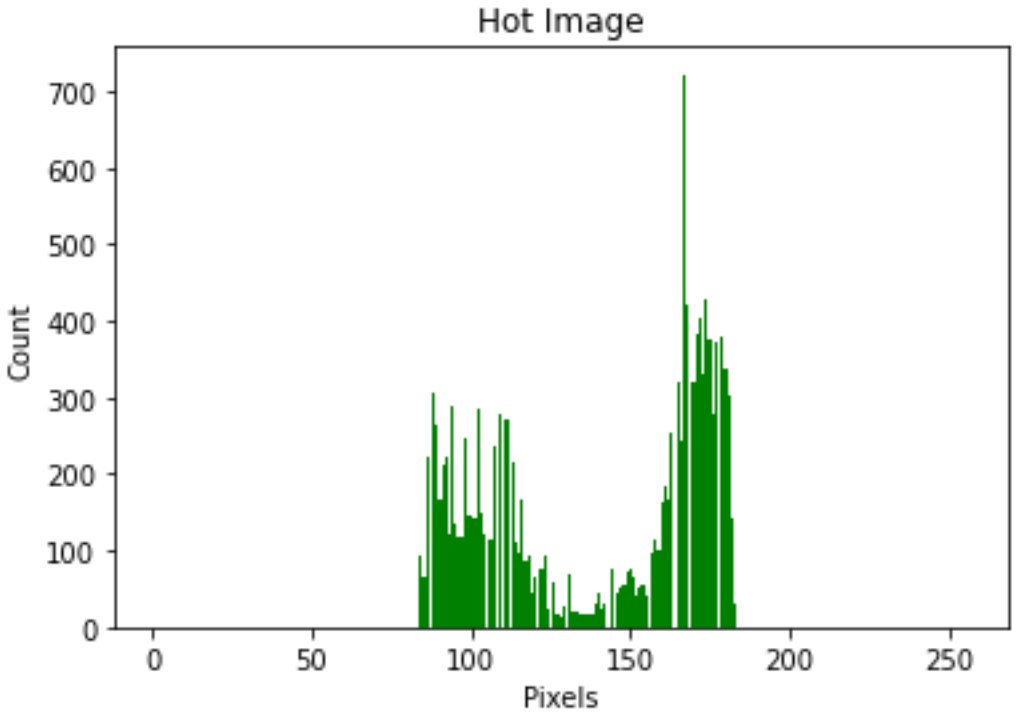
Histogram Features	Textural Features
Mean	Dissimilarity
Entropy	Correlation
Skewness	Energy
Gray scale pixel count	Homogeneity
	Contrast
	Angular Second Moment (ASM)

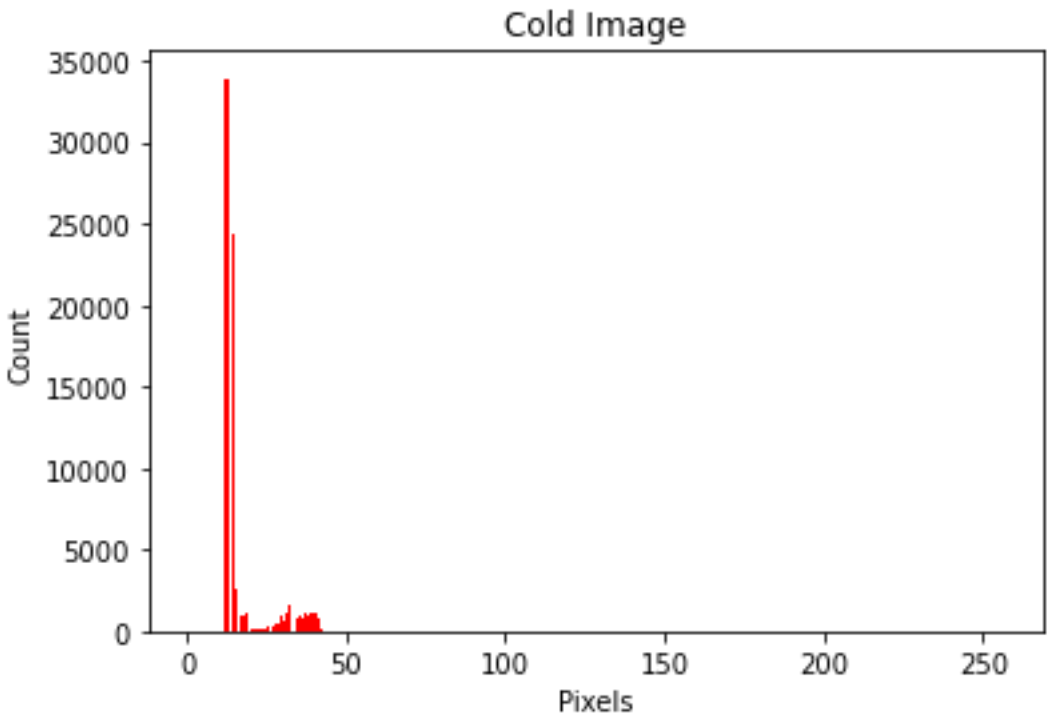


Implementation (Cont'd)

Feature Extraction

- Distribution of gray pixels in an image





Implementation (Cont'd)

Machine Learning

- Training 70% and Testing 30%

ML Classifiers

- Decision Tree Classifier
- Random Forest Classifier
- Ada Boost Classifier
- Gradient Boosting Trees Classifier

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score



Implementation (Cont'd)

Hot or Cold classification

- First-order feature extraction is used for hot and cold classification
- Hence, gray color pixels are considered to be in the middle of the specified range and are considered as the region of interest
- The rest of the pixels are converted to black, i.e. Zero
- Here only the pixels of gray color is taken into consideration, which is a non-zero count
- Otsu automatic thresholding function is used for optimal threshold value



Implementation (Cont'd)

Hot or Cold classification

- The cold class is considered as 0 and the hot class is considered as 1.

	Hot images	Cold images
Image Count	360	180
Number of classes	4	2
Classification	Multi-classification	Binary classification
Classes	Cooling fan failure	Stator fault (1-Phase)
	Stator fault (3-Phase)	
	Stator fault (2-Phase)	Healthy
	Rotor fault	
Training (70%)	252	108
Testing (30%)	126	54

Implementation (Cont'd)

ROI Classification – Thresholding Functions

- Otsu automatic thresholding function is used for finding ground truth
- Distinguish the foreground from the background
- Cluster-based image thresholding

ROI Classification – Ground truth

- Accurate segmentation or partition of an image
- Region of interest is enclosed with a rectangular bounding box based on ground truth



Implementation (Cont'd)

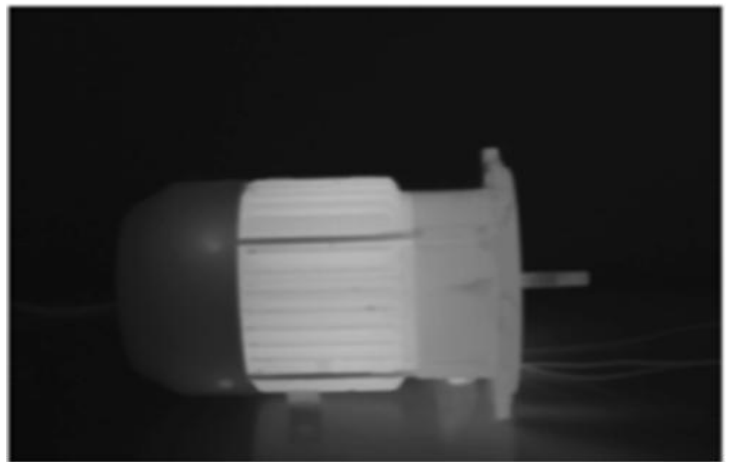
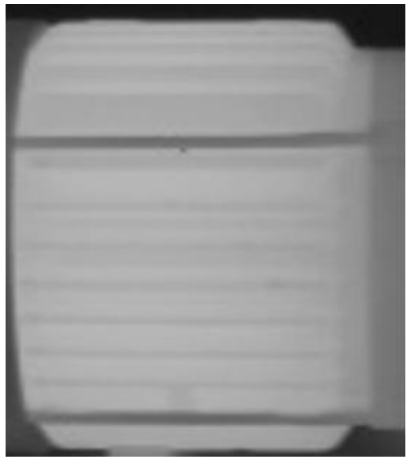
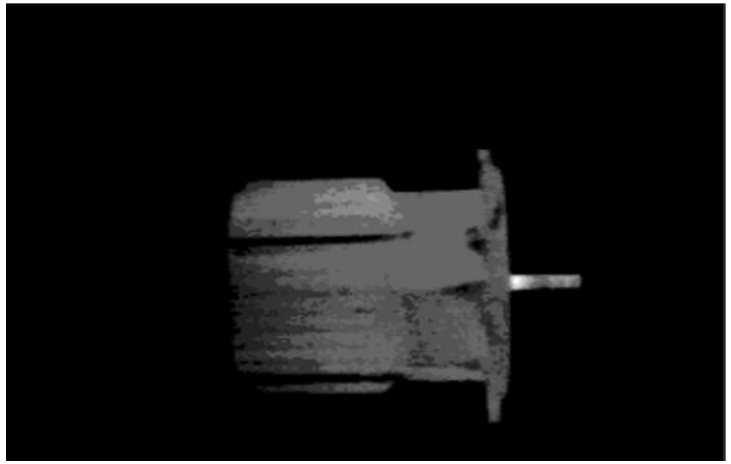
ROI Classification – Sliding Window Technique

- The sliding window technique is used to find the region of interest in an image.
- It consists of the window size, step size, also known as stride value.
- The window size is considered as (9 x 16) and the stride value is chosen as 1
- Ground truth is compared to all the obtained images from sliding window from which Region of interest is detected



Implementation (Cont'd)

ROI – Hot and Cold Images


Implementation (Cont'd)

Fault Classification

- The first and second-order features are extracted from predicted RoI images and faults are classified using previously trained models
- Models are used to compare and evaluate the actual and predicted conditions of each image.
- The prediction results are used to calculate overall accuracy.

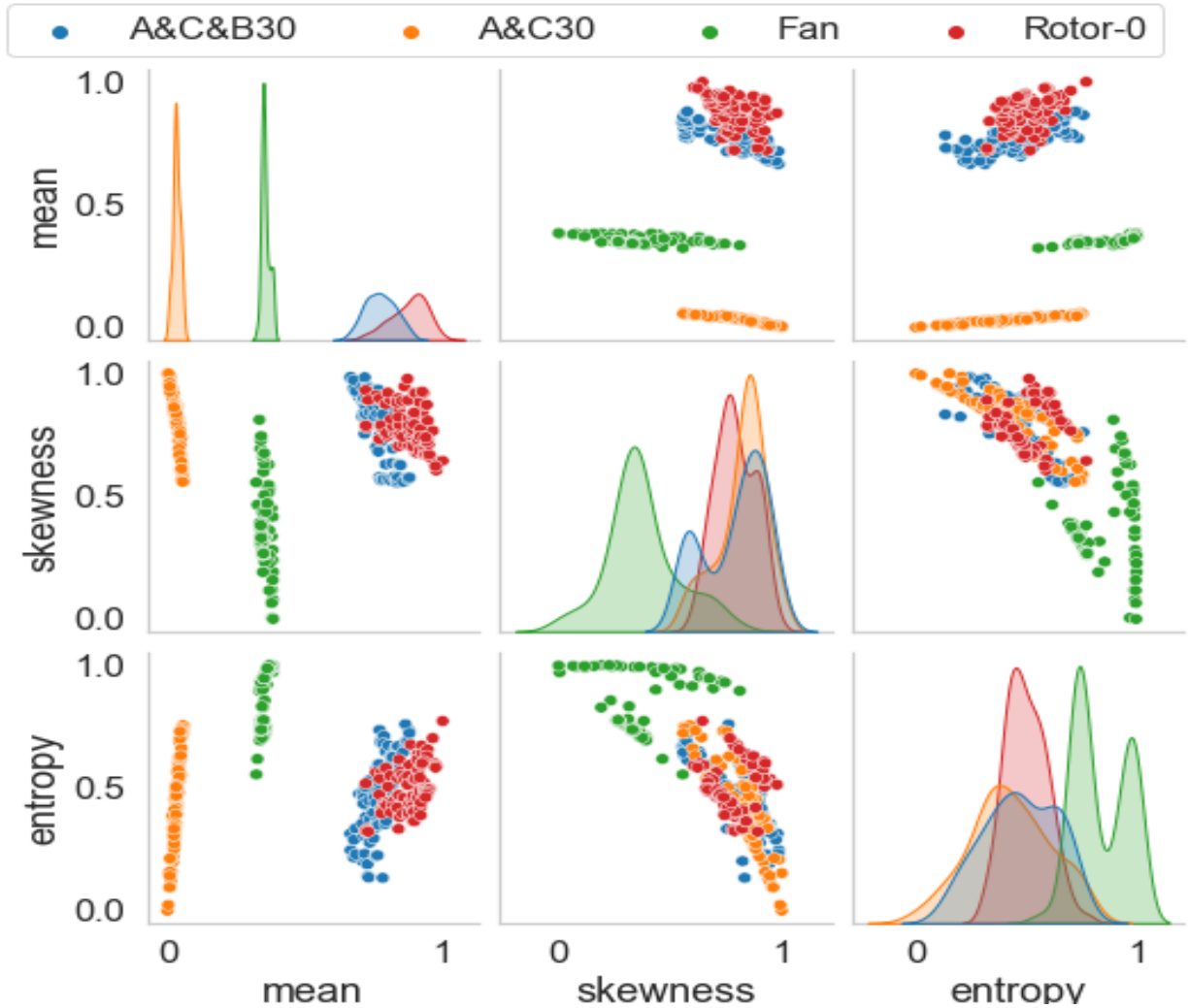


Results

Features of hot images

First-Order / Histogram features

- Mean
- Skewness
- Entropy

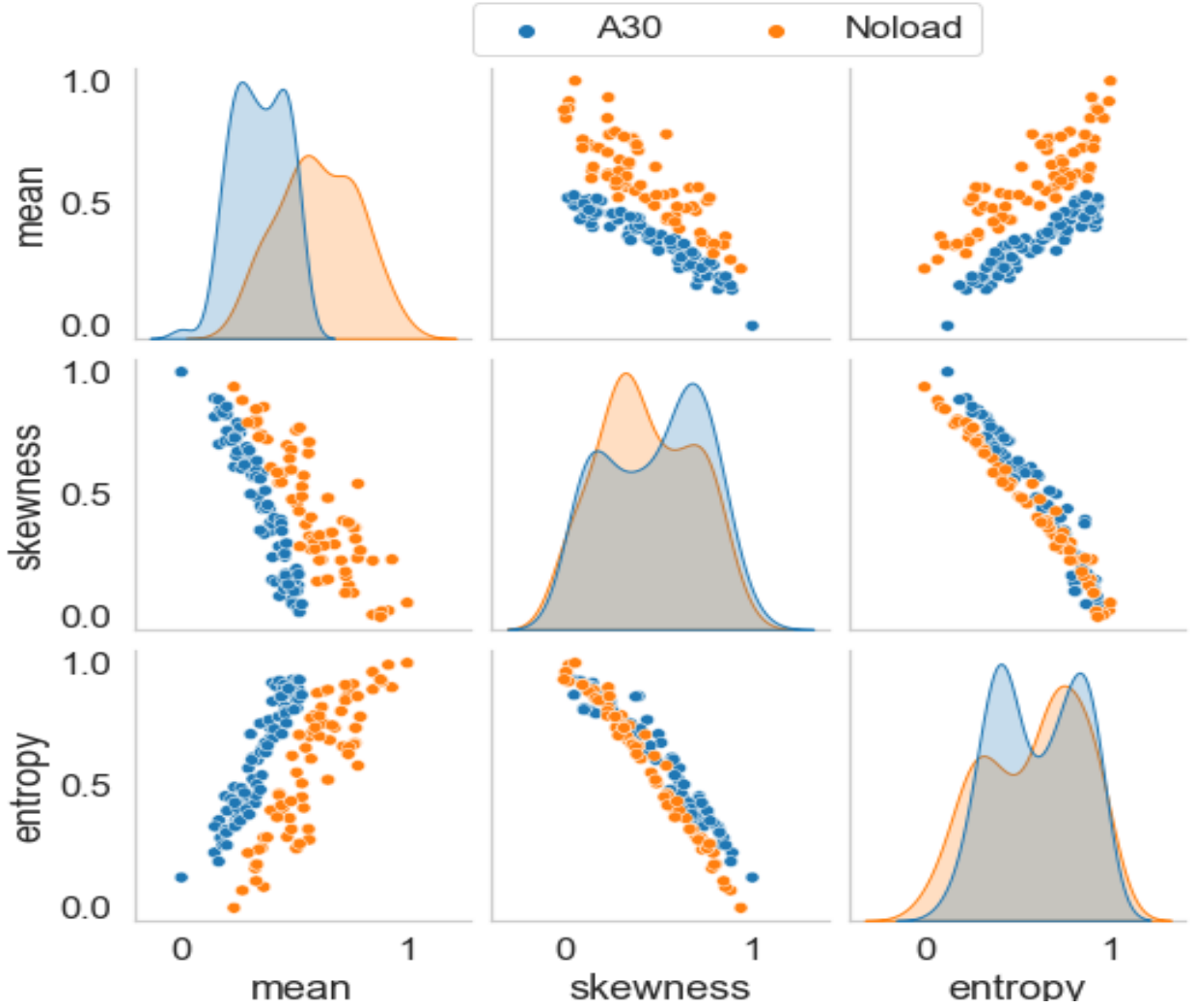


Results (Cont'd)

Features of cold images

First-Order / Histogram features

- Mean
- Skewness
- Entropy



Results (Cont'd)

- **Hot or cold classification**

ML Classifiers	Accuracy in Percentage (%)	F1 Score
Decision Tree Classifier	100	1.00
Random Forest Classifier	100	1.00
Ada Boost Classifier	100	1.00
Gradient Boosting Trees Classifier	100	1.00



Results (Cont'd)

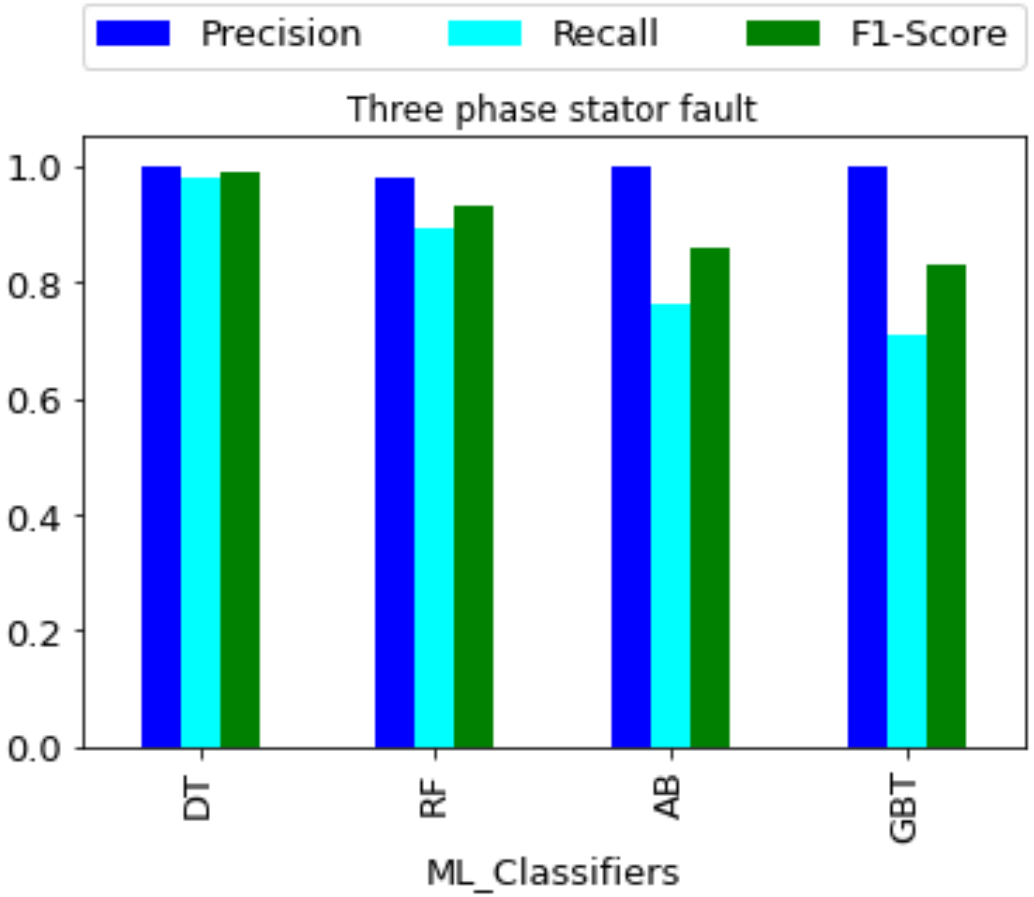
Validation Set

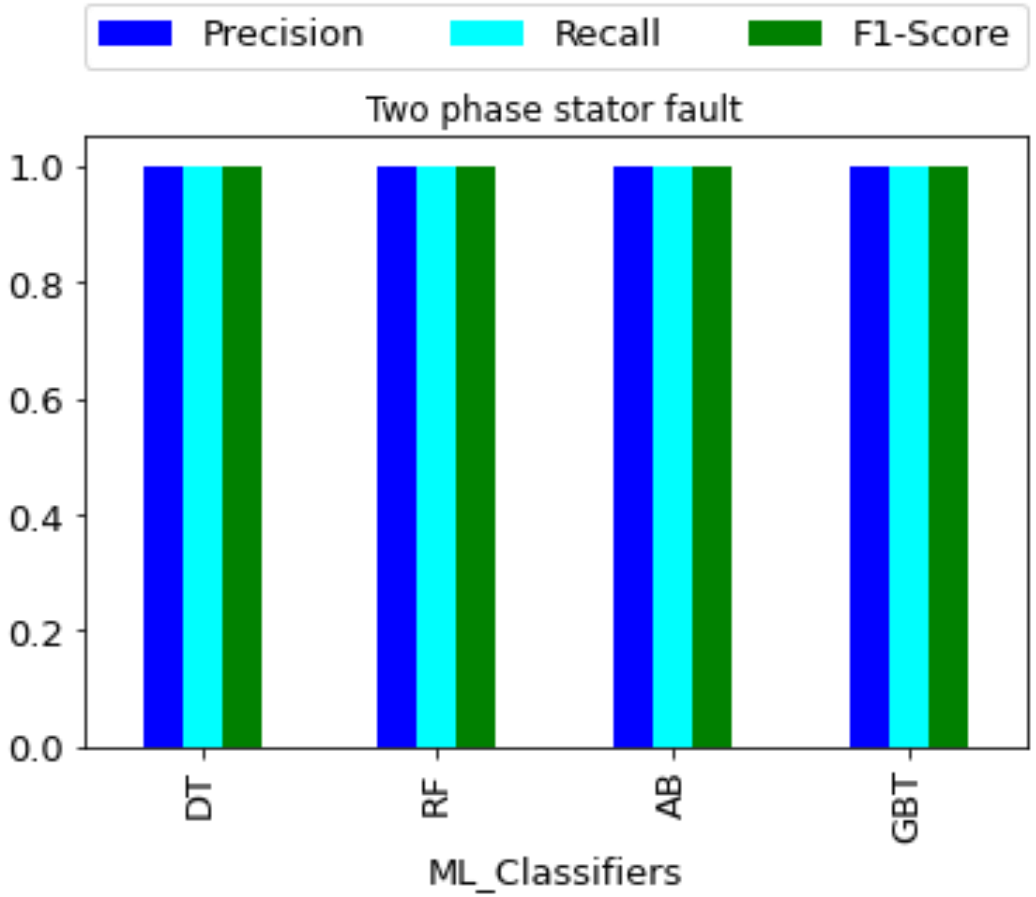
- A validation set includes unseen data and used data selected at random
- 180 hot images and 90 cold images are considered for analysis
- Trained classification models are used for final fault detection



Results (Cont'd)

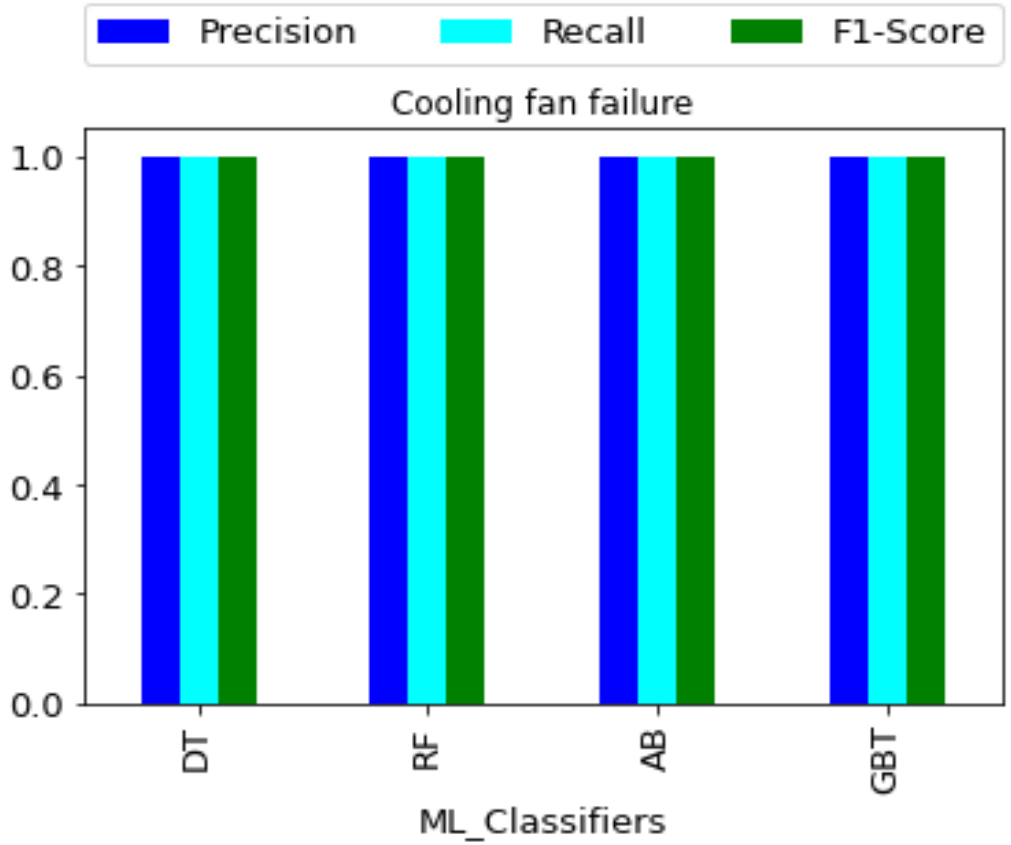
Evaluation Metrics – Hot

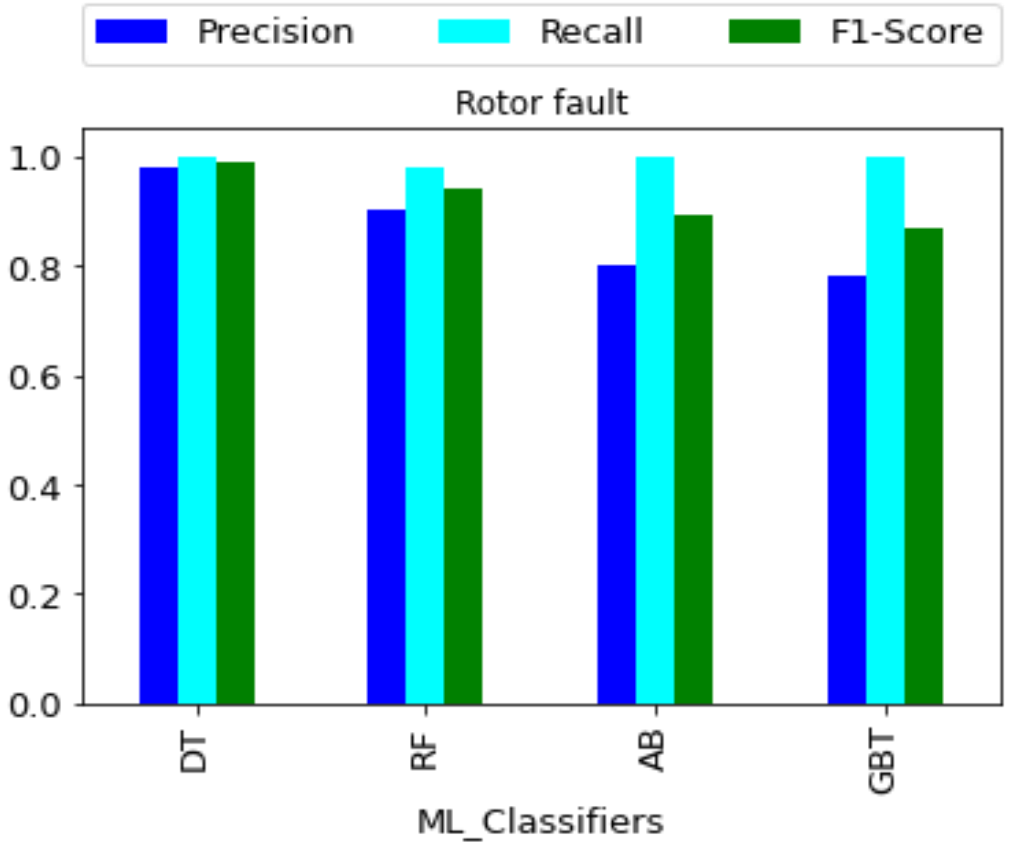




Results (Cont'd)

Evaluation Metrics – Hot





Results (Cont'd)

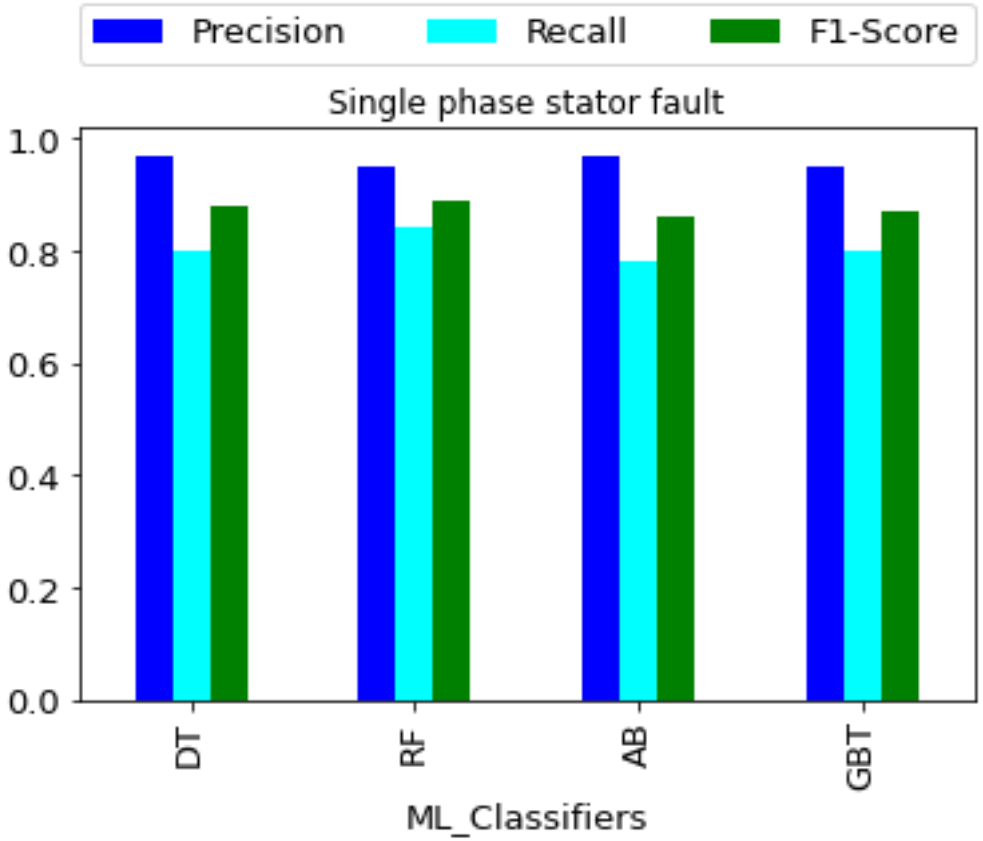
- **Evaluation metrics for hot class**

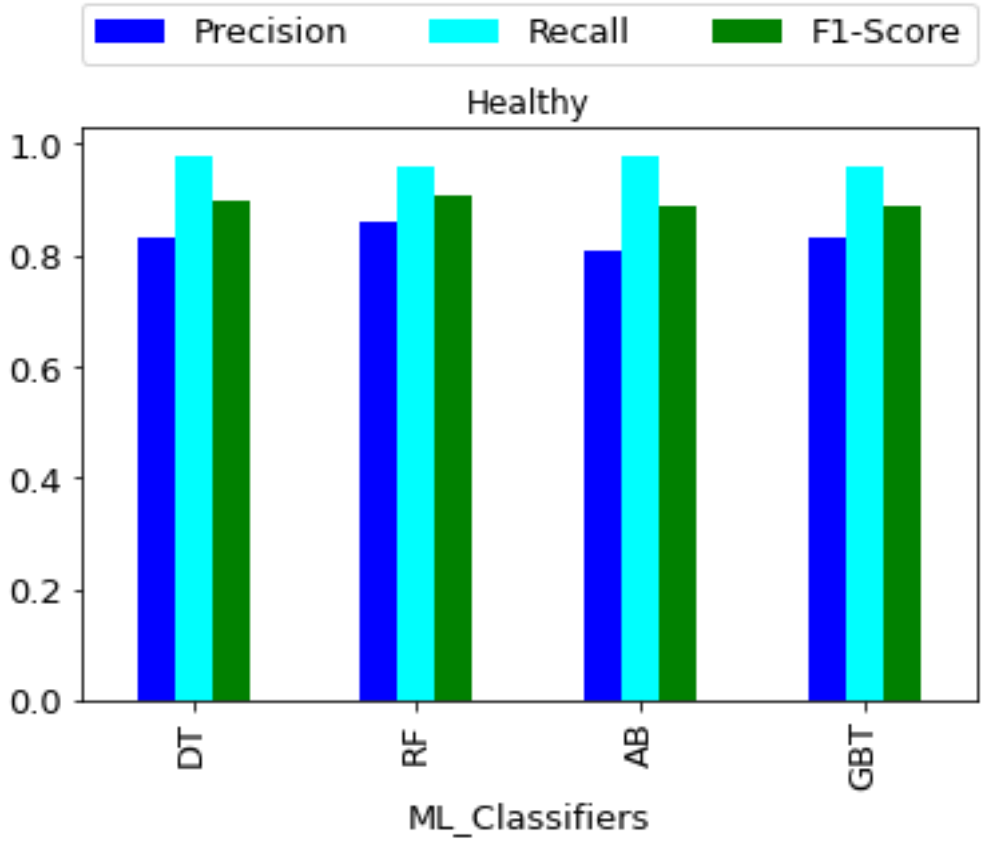
ML Classifiers	Accuracy in Percentage (%)	F1 Score
Gradient Boosting Trees Classifier	92.8	0.92
Ada Boost Classifier	93.8	0.93
Decision Tree Classifier	93.9	0.94
Random Forest Classifier	**97.2**	**0.97**



Results (Cont'd)

Evaluation Metrics – Cold





Results (Cont'd)

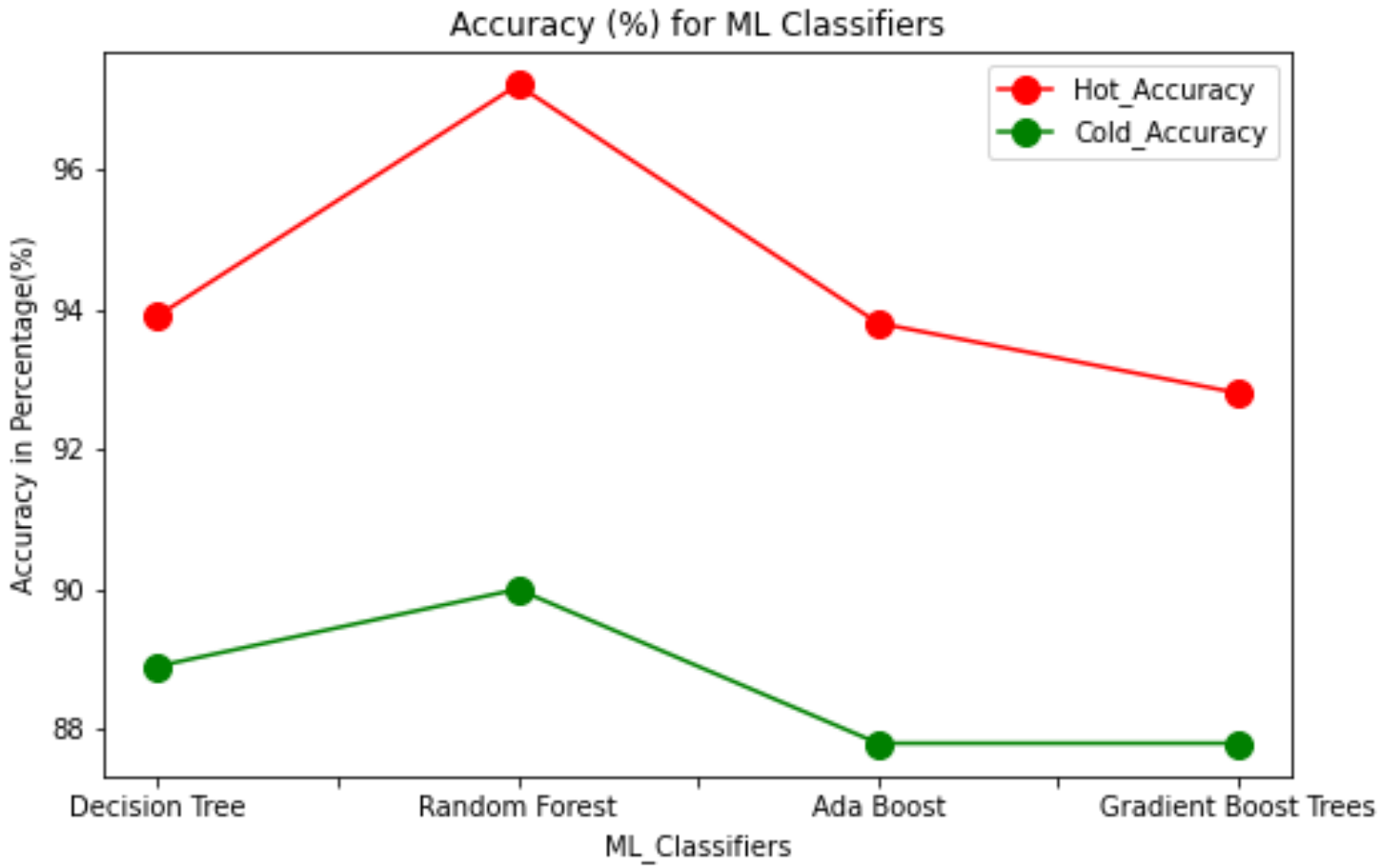
- **Evaluation metrics for cold class**

ML Classifiers	Accuracy in (%)	Precision	Recall	F1 Score
Ada Boost Classifier	87.8	0.97	0.81	0.89
Gradient Boosting Trees Classifier	87.8	0.96	0.82	0.89
Decision Tree Classifier	88.9	0.97	0.83	0.89
Random Forest Classifier	**90**	**0.96**	**0.86**	**0.90**



Results (Cont'd)

- **Overall accuracy for hot and cold classes**



Conclusion

- The project provides a condition monitoring system for fault detection in induction motor
- The main focus is to detect the internal faults present in an induction motor using infrared thermal images
- Classification of images into hot and cold helps in locating the specific faults
- The region of interest is found accurately using a sliding window technique
- Faults are classified with an overall accuracy of 97.2% for hot conditions and 90% for cold conditions



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THANK YOU